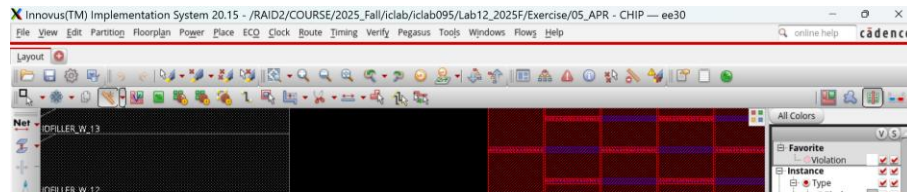


Report

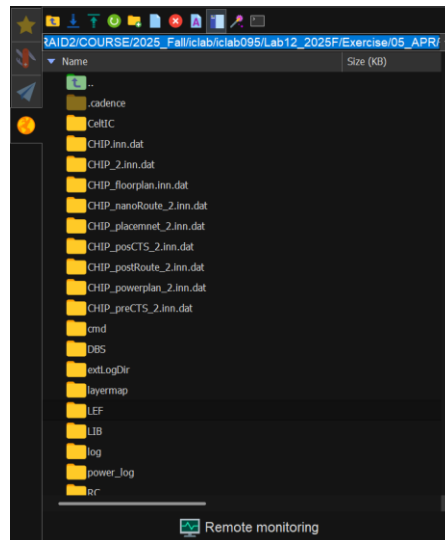
1. server info



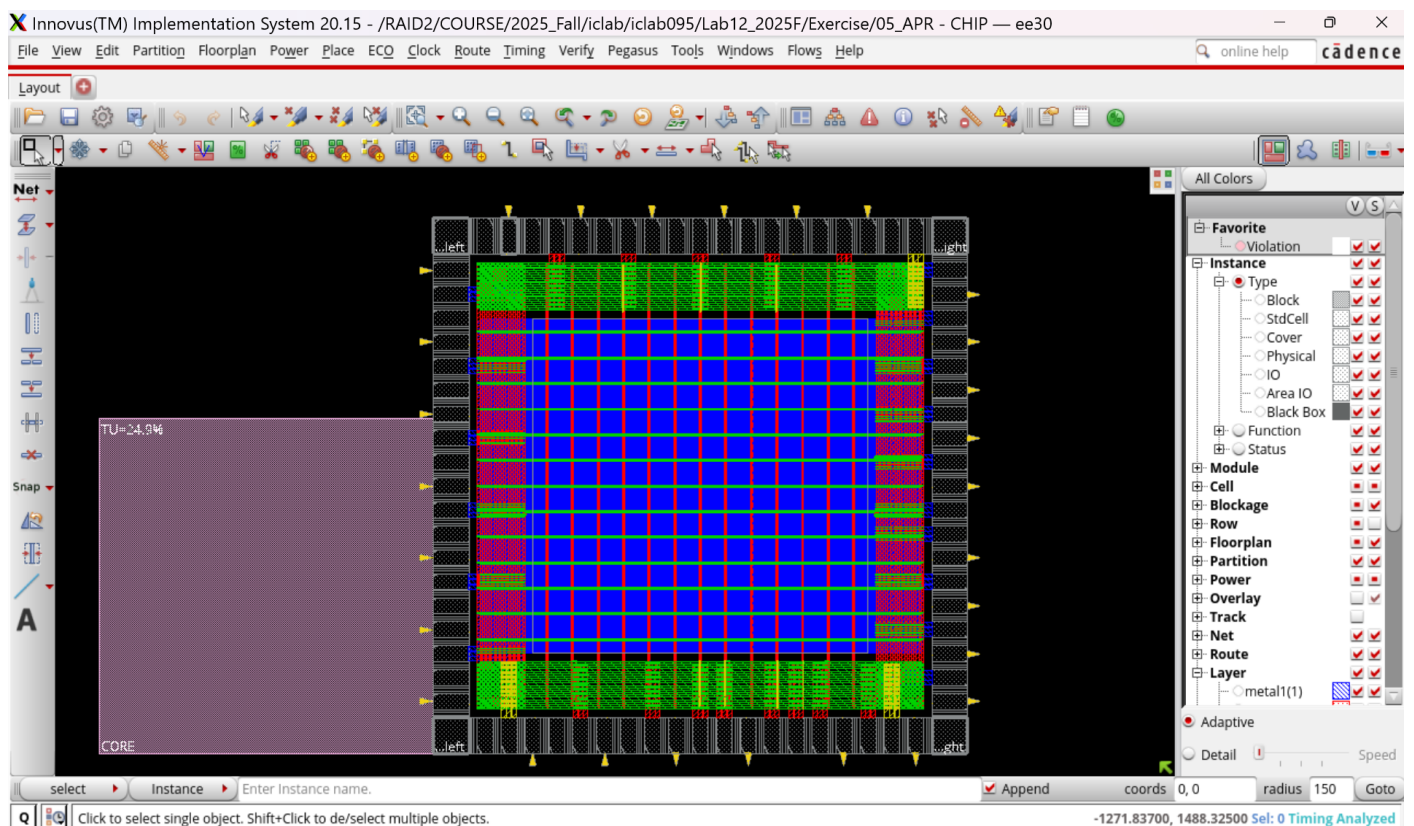
2. server info

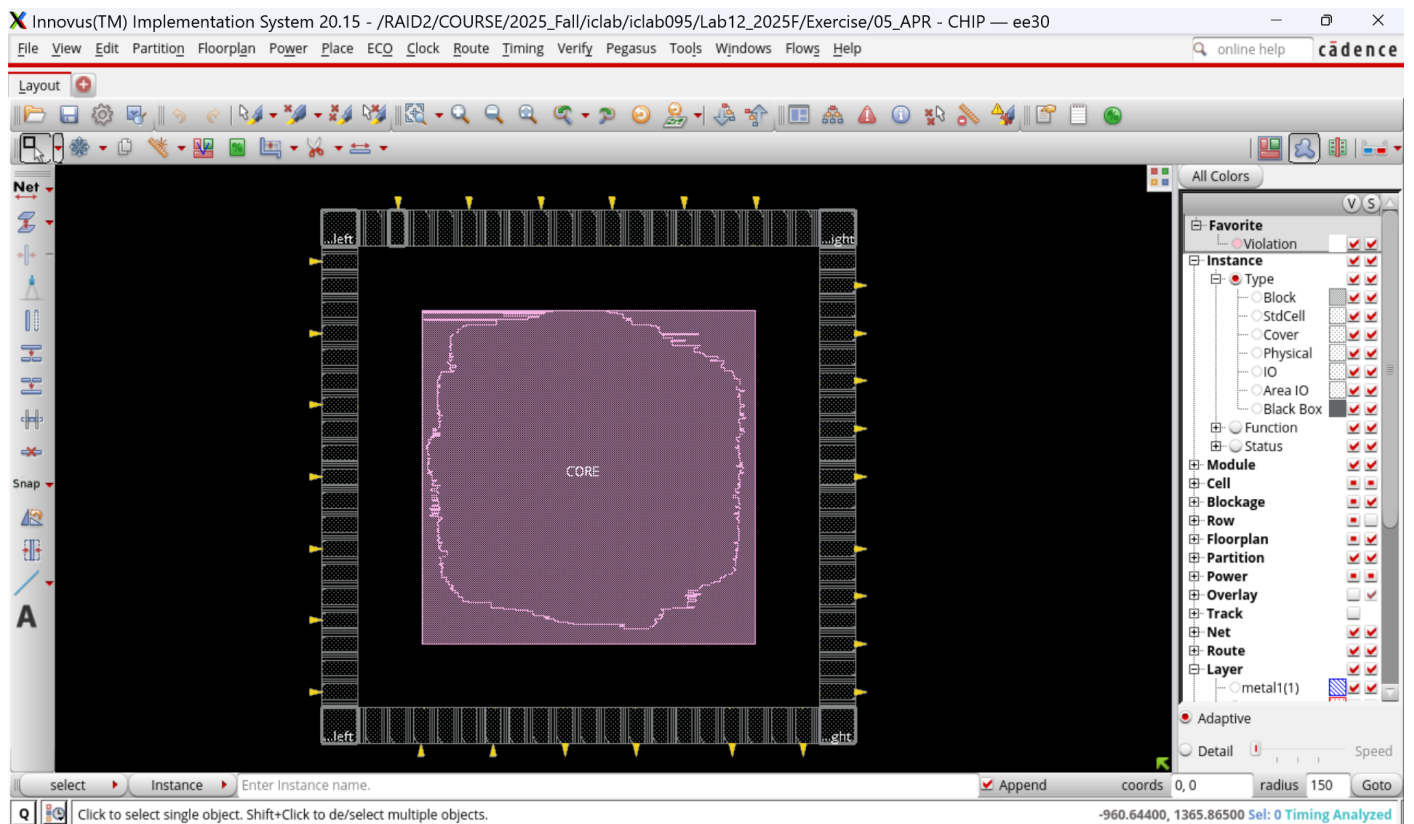
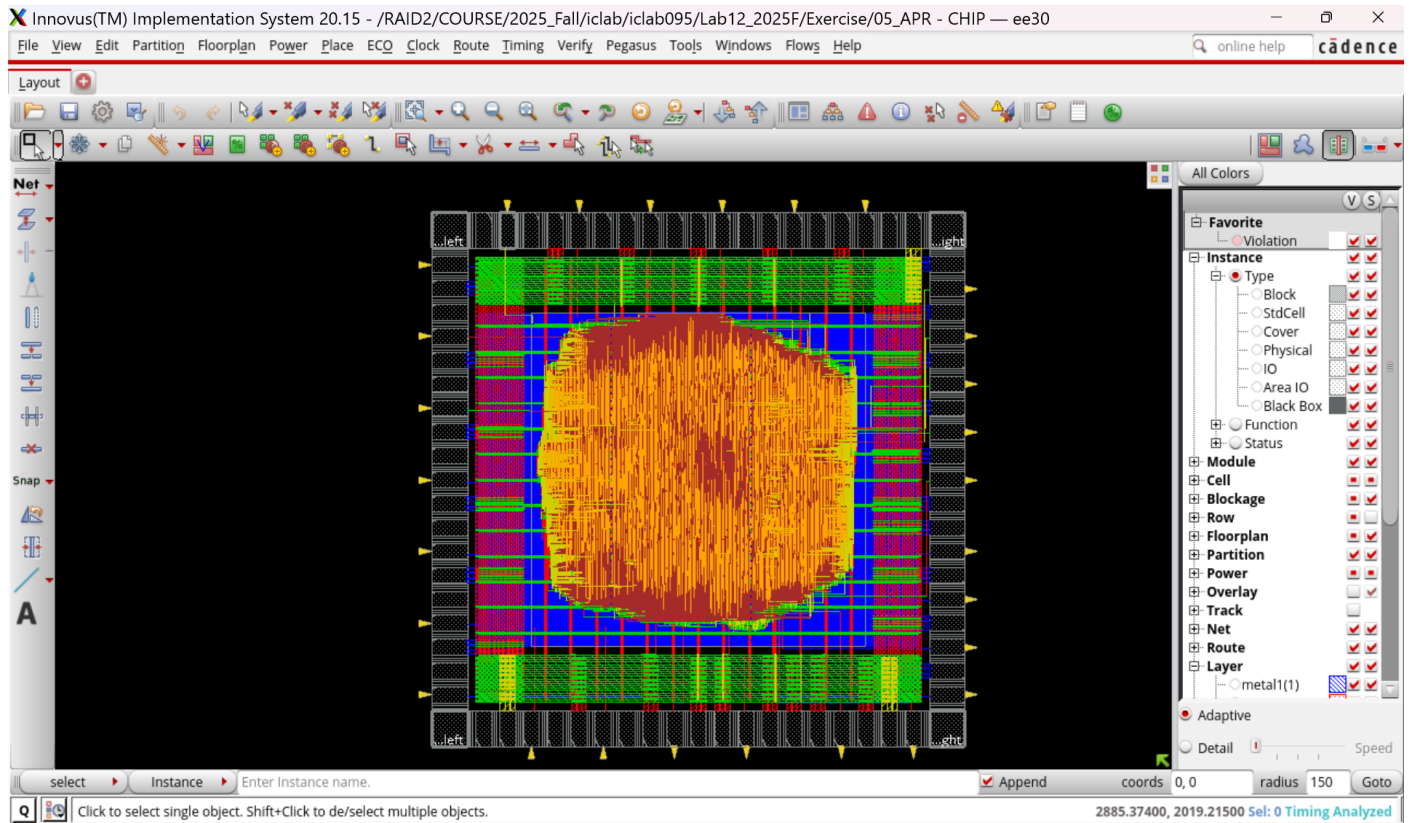
```
21:16 iclab095@ee30[~/Lab12_2025F/Exercise]$ cd 05_APR/  
21:16 iclab095@ee30[~/Lab12_2025F/Exercise/05_APR]$ setenv OA_HOME /RAID2/cad/cadence/INNOVUS  
/INNOVUS_20.15.000/share/oa  
21:17 iclab095@ee30[~/Lab12_2025F/Exercise/05_APR]$ innovus
```

3. server info

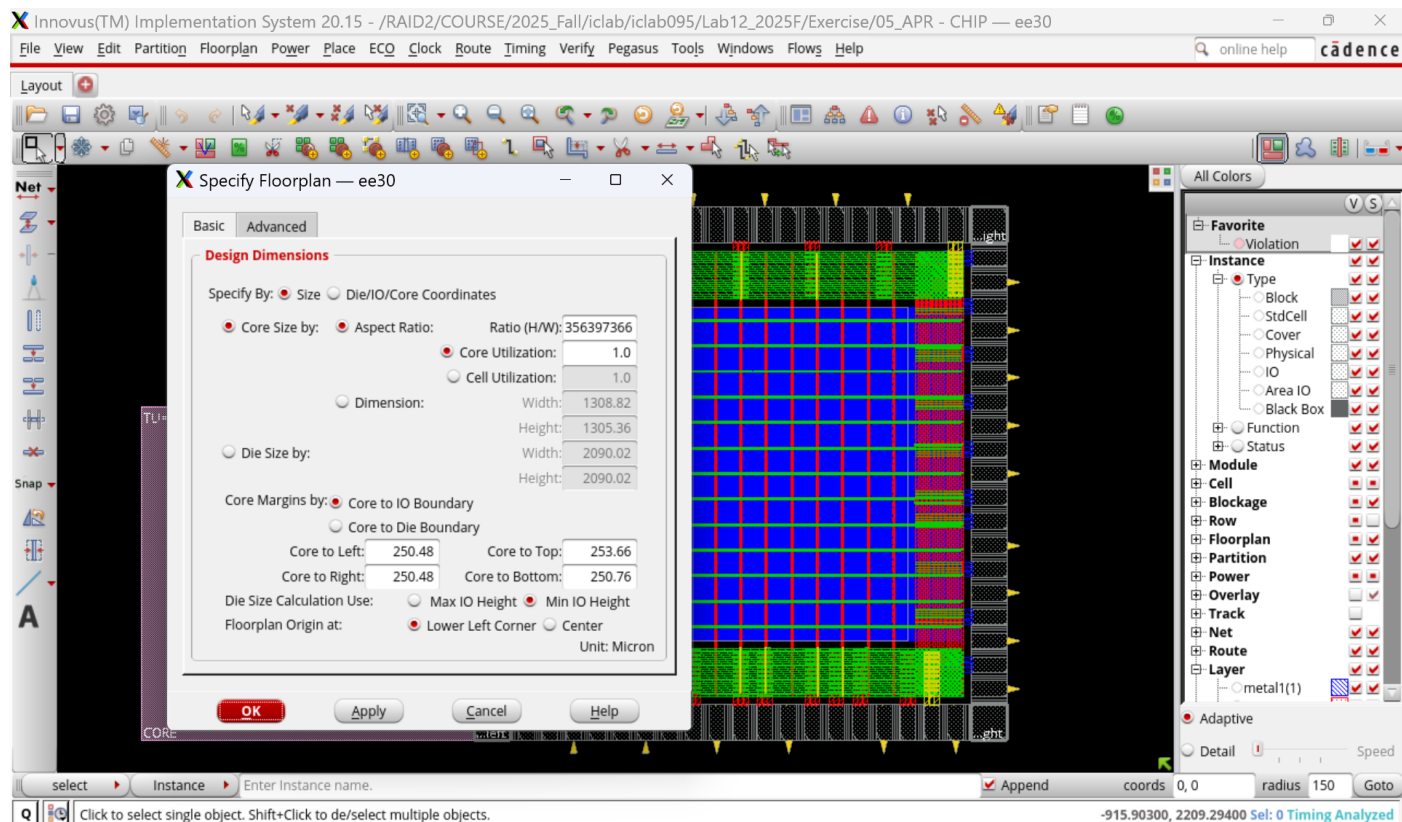


1. Chip Layout View



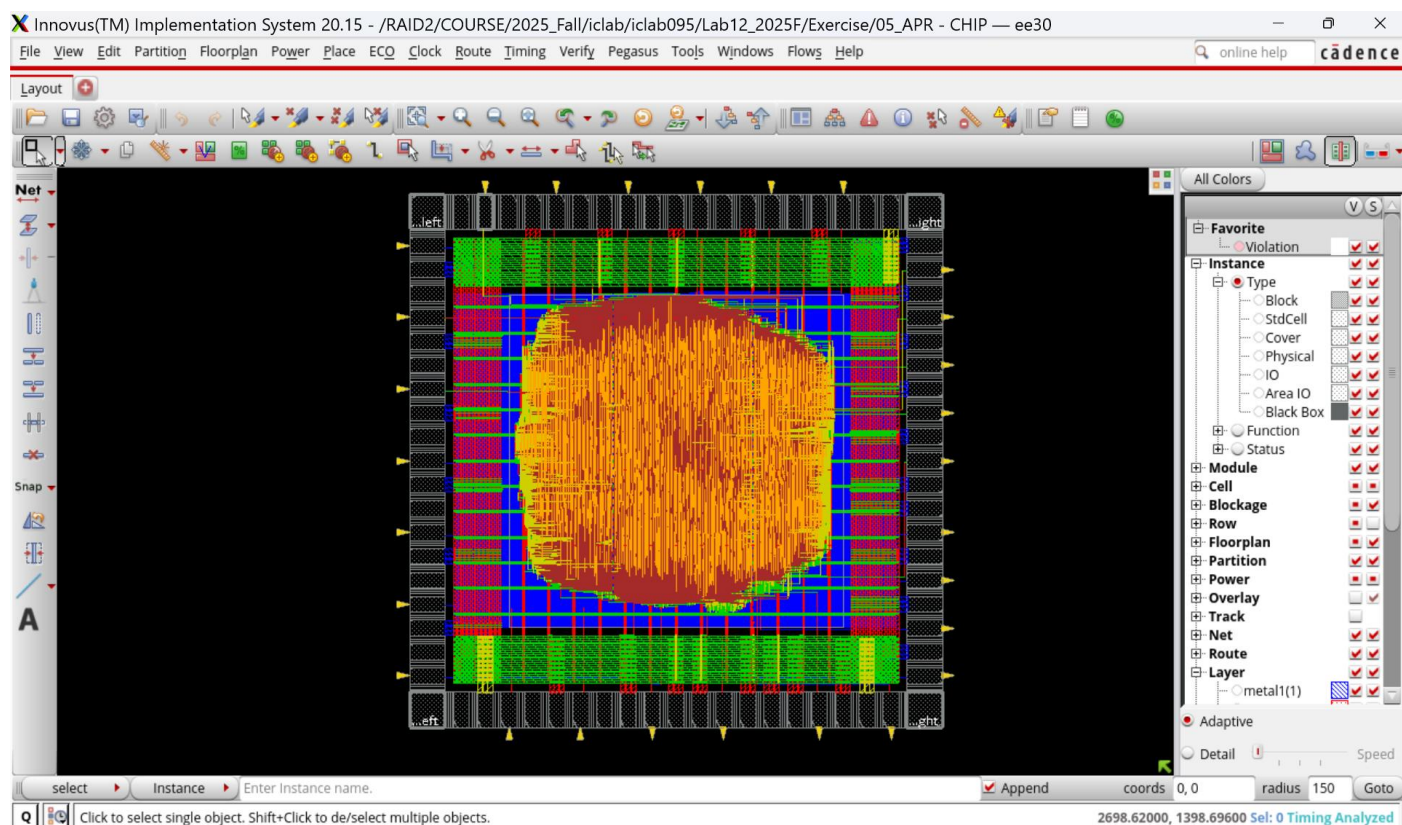


2. Core to IO boundary



3. Core Ring

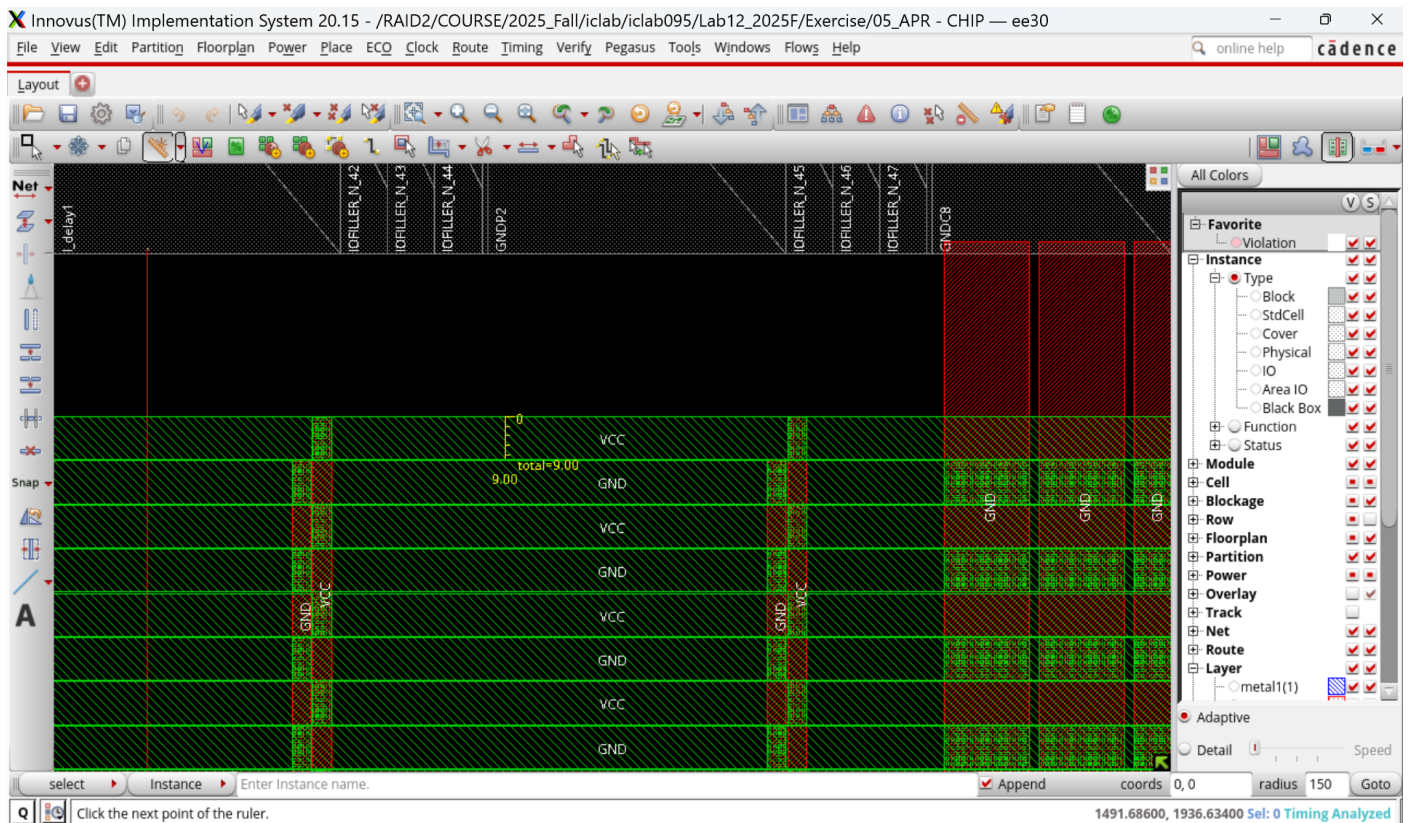
Overall Power Ring



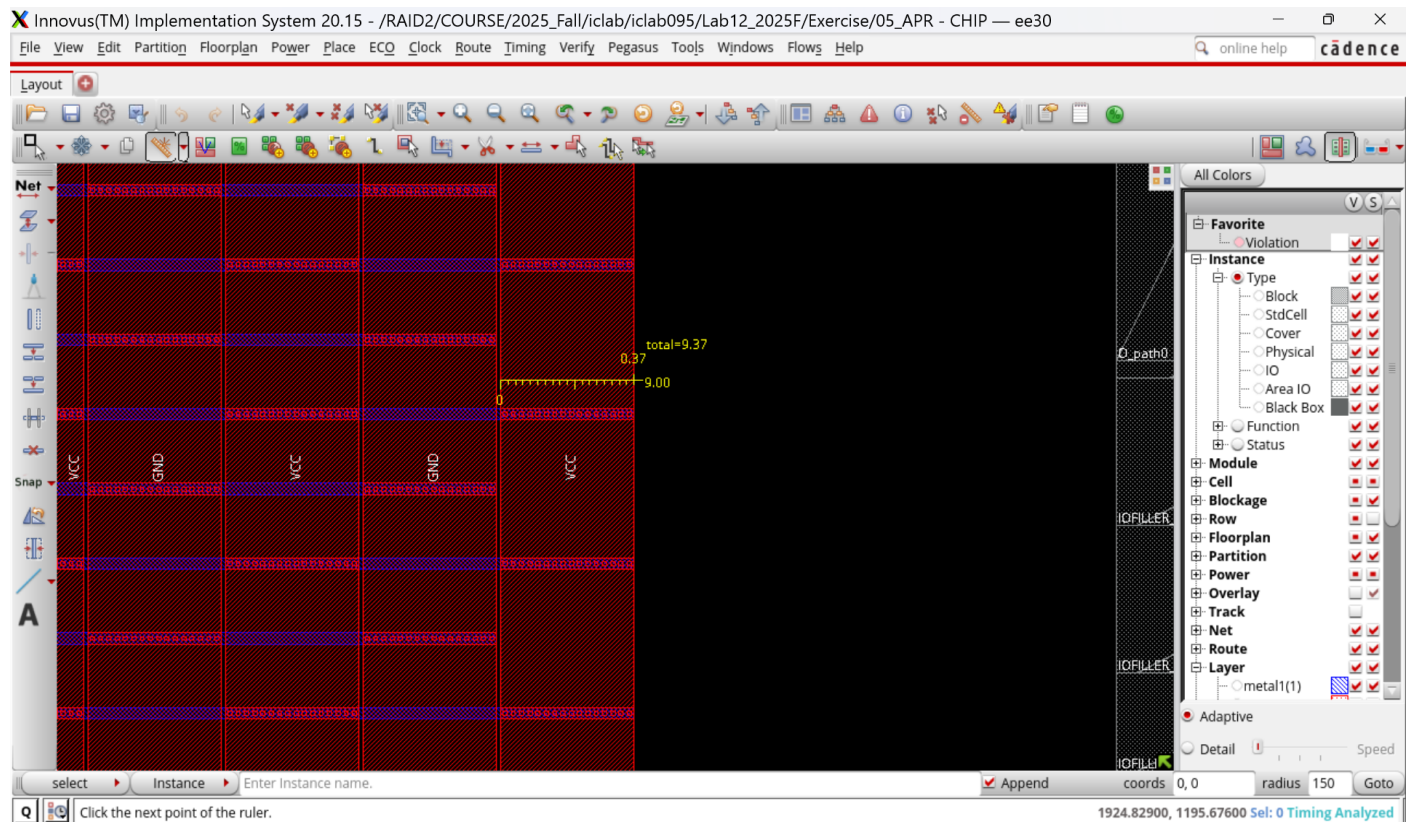
Overall Power Ring: wire group, interleaving, 10 pairs, width 9.



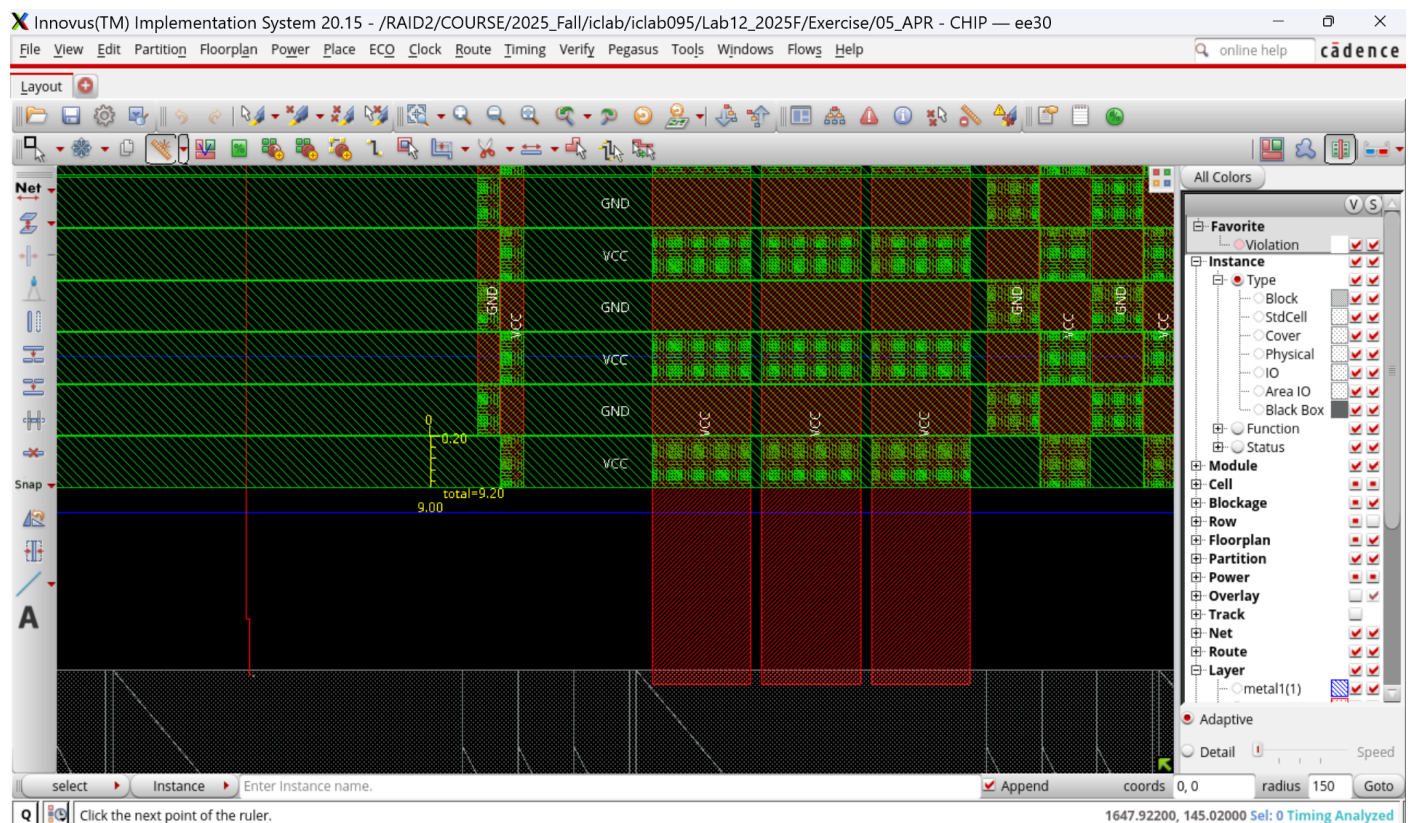
Top Power Ring: width=9



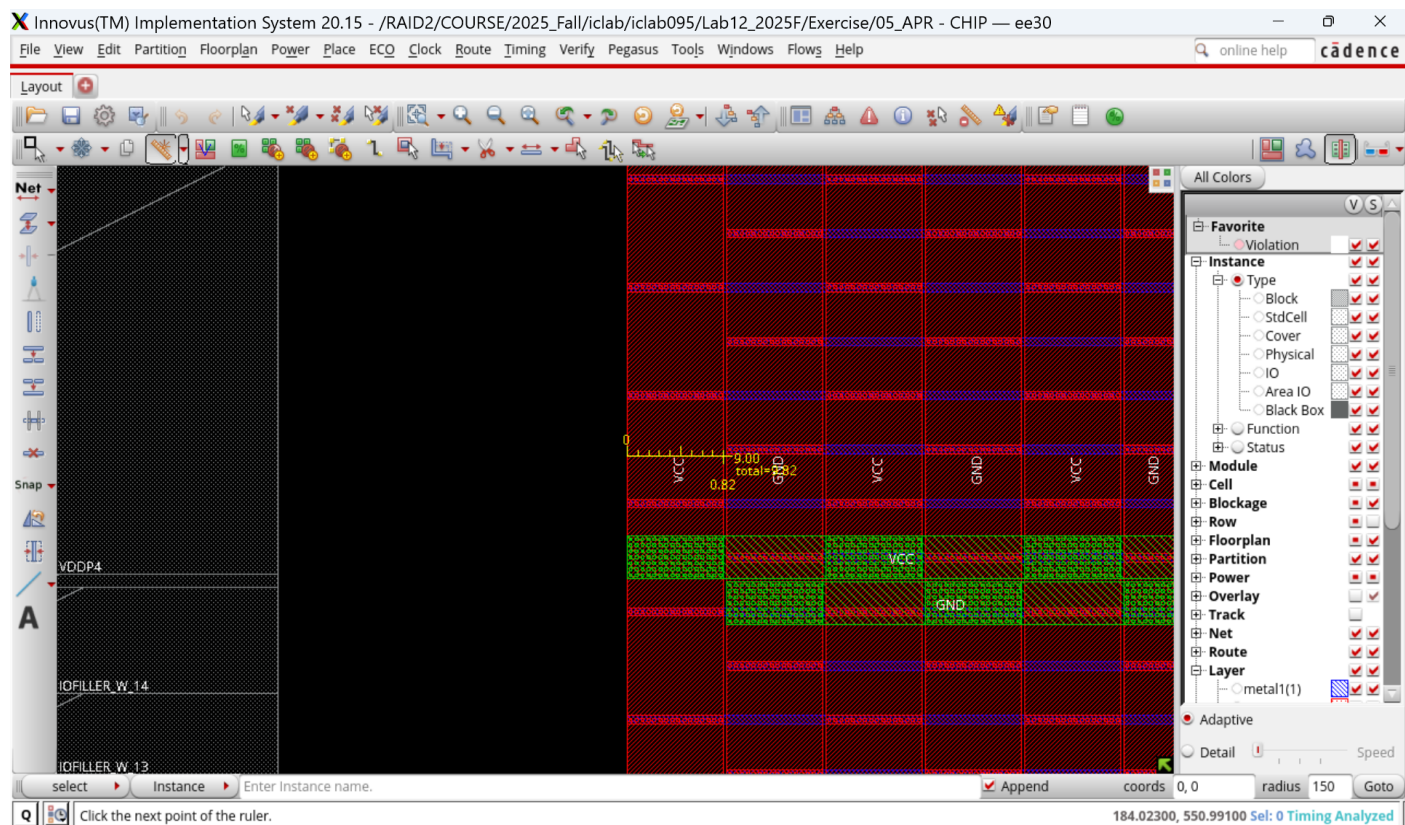
Right Power Ring: width=9



Bottom Power Ring: width=9



Left Power Ring: width=9



(Width settings are all 9)

4. Post-Route setup time analysis :

ee30.si2.iee.nycu.edu.tw (iclab095)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

/RAID2/COURSE/2025_Fall/iclab/iclab095/Lab12_2025F/Exercise/05_APF

Name Size (KB)

- LEF
- LIB
- log
- power_log
- RC
- summaryReport
- timingReports
- work
- .qor_metric.tcl 1
- .timing_file_154986.tif.gz 1 902
- .timing_file_173074.tif.gz 1 903
- .timing_file_239865.tif.gz 1 902
- 00_combine 1
- 01_setenv 1
- 09_clean_up 1
- cellIDMap 1
- CHIP.CCOPC.spec 2
- CHIP.conn.rpt 1
- CHIP.conn.rpt.old 1
- CHIP.drc.rpt 1
- CHIP.drc.rpt.old 1

Remote monitoring

Follow terminal folder

Loading remote monitoring, please wait...

*** Done Building Timing Graph (cpu=0:00:12.0 real=0:00:12.0 totSessionCpu=0:01:55 mem=2342.4M)

timeDesign Summary

Setup views included:
av_func_mode_max

Setup mode	all	reg2reg	default
WNS (ns):	0.011	0.098	0.011
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	1415	688	727

DRVs	Real		Total	
	Nr nets(terms)	Worst Vio	Nr nets(terms)	
max_cap	0 (0)	0.000	0 (0)	
max_tran	0 (0)	0.000	0 (0)	
max_fanout	0 (0)	0	0 (0)	
max_length	0 (0)	0	0 (0)	

Density: 33.397%
(126.299% with Fillers)

Total number of glitch violations: 0

Reported timing to dir timingReports

Total CPU time: 93.08 sec

Total Real time: 95.0 sec

Total Memory Usage: 2333.964844 Mbytes

Reset AAE Options

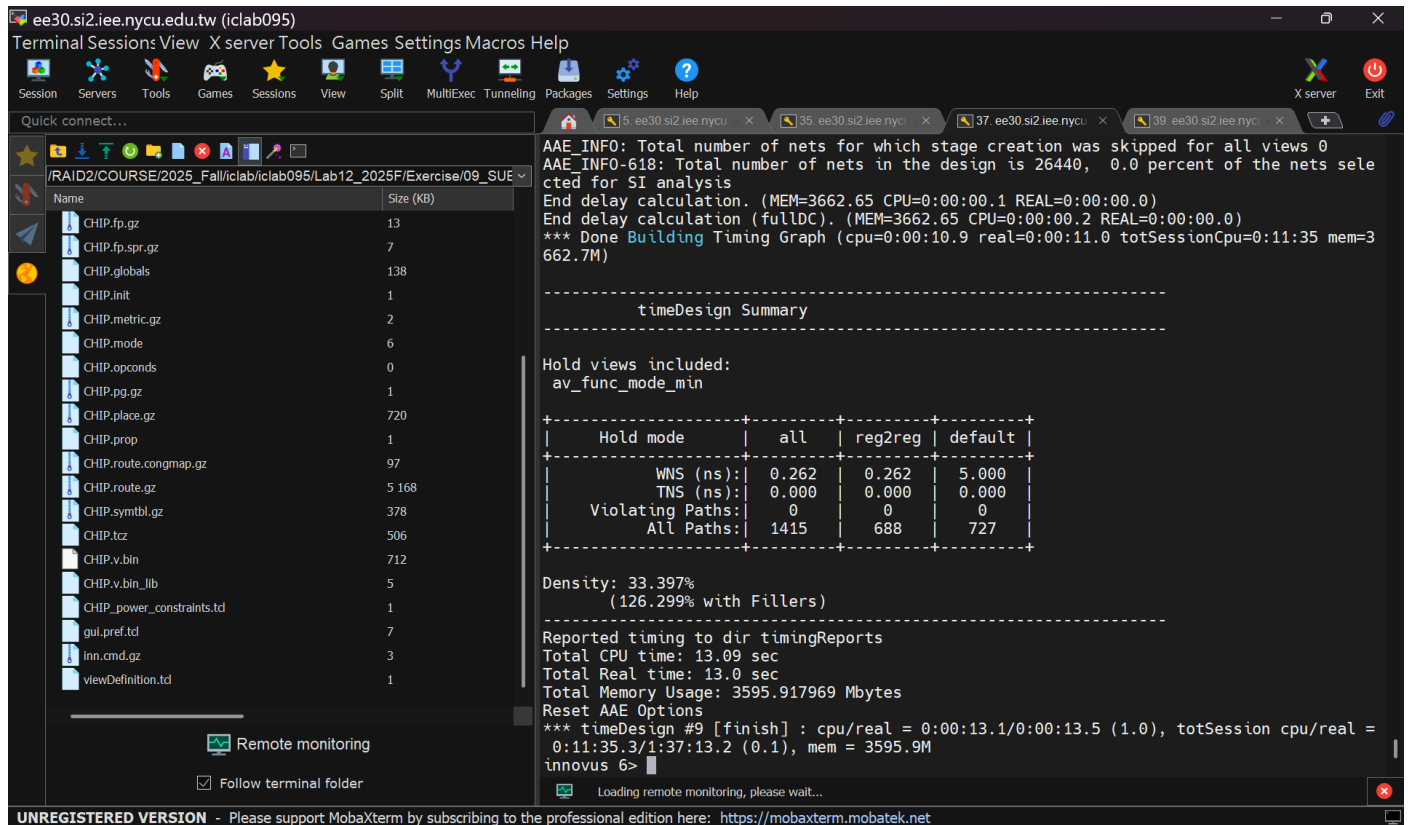
*** timeDesign #1 [finish] : cpu/real = 0:01:33.1/0:01:35.1 (1.0), totSession cpu/real = 0:01:56

.9/0:03:01.2 (0.6), mem = 2334.0M

innovus S>

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

5. Post-Route hold time analysis



The screenshot shows the MobaXterm interface with a terminal window displaying the results of a post-route hold time analysis. The left sidebar shows a file explorer with a directory structure for a course. The main terminal window displays the following output:

```
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 26440, 0.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=3662.65 CPU=0:00:00.1 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=3662.65 CPU=0:00:00.2 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:10.9 real=0:00:11.0 totSessionCpu=0:11:35 mem=3662.7M)

-----
timeDesign Summary
-----

Hold views included:
av_func_mode_min

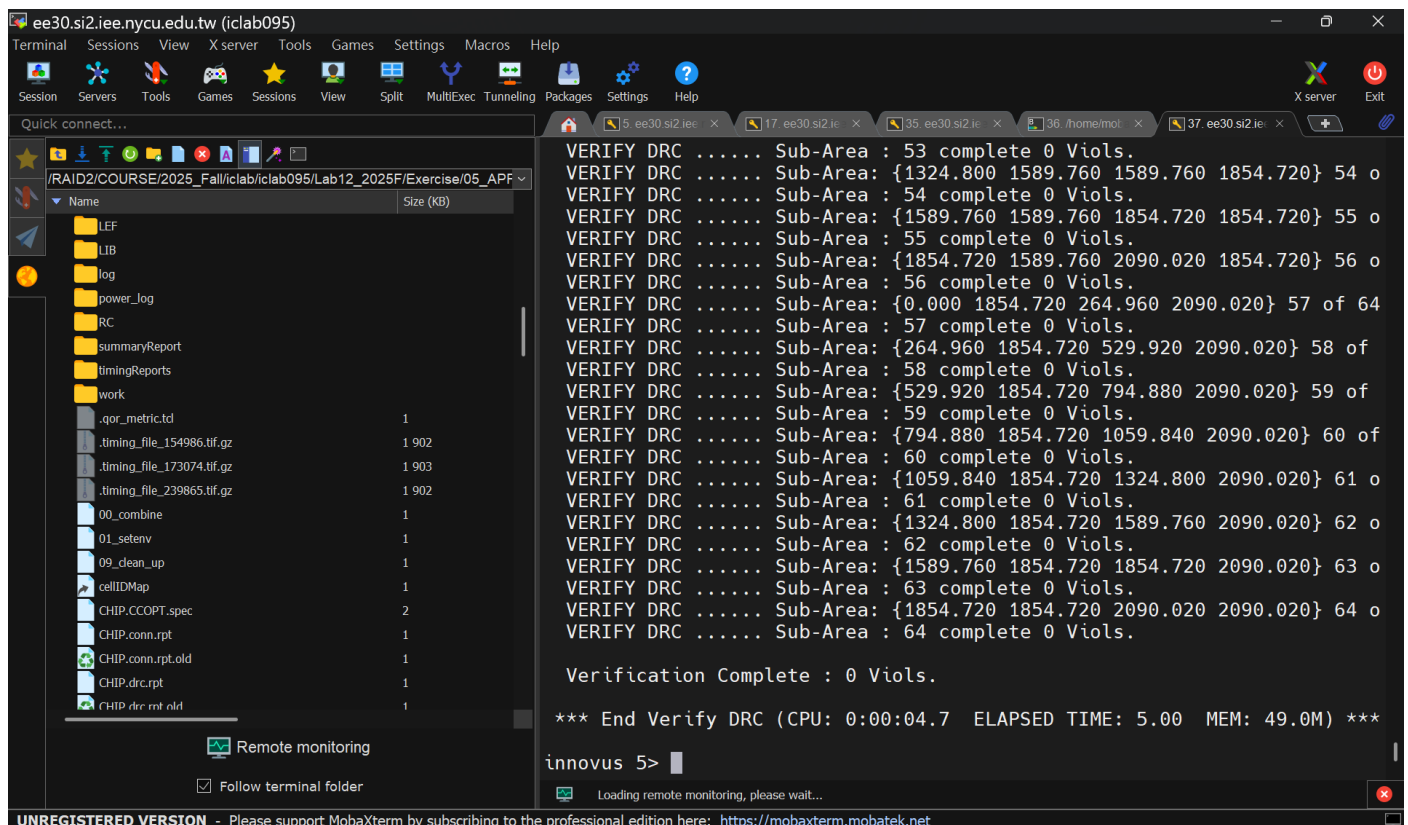
+-----+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns) | 0.262 | 0.262 | 5.000 |
| TNS (ns) | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 1415 | 688 | 727 |
+-----+-----+-----+-----+

Density: 33.397%
(126.299% with Fillers)

Reported timing to dir timingReports
Total CPU time: 13.09 sec
Total Real time: 13.0 sec
Total Memory Usage: 3595.917969 Mbytes
Reset AAE Options
*** timeDesign #9 [finish] : cpu/real = 0:00:13.1/0:00:13.5 (1.0), totSession cpu/real = 0:11:35.3/1:37:13.2 (0.1), mem = 3595.9M
innovus 6>
```

The bottom of the terminal window shows the status "UNREGISTERED VERSION" and a link to the professional edition of MobaXterm.

6. DRC result



The screenshot shows the MobaXterm interface with a terminal window displaying the results of a Design Rule Check (DRC). The left sidebar shows a file explorer with a directory structure for a course. The main terminal window displays the following output:

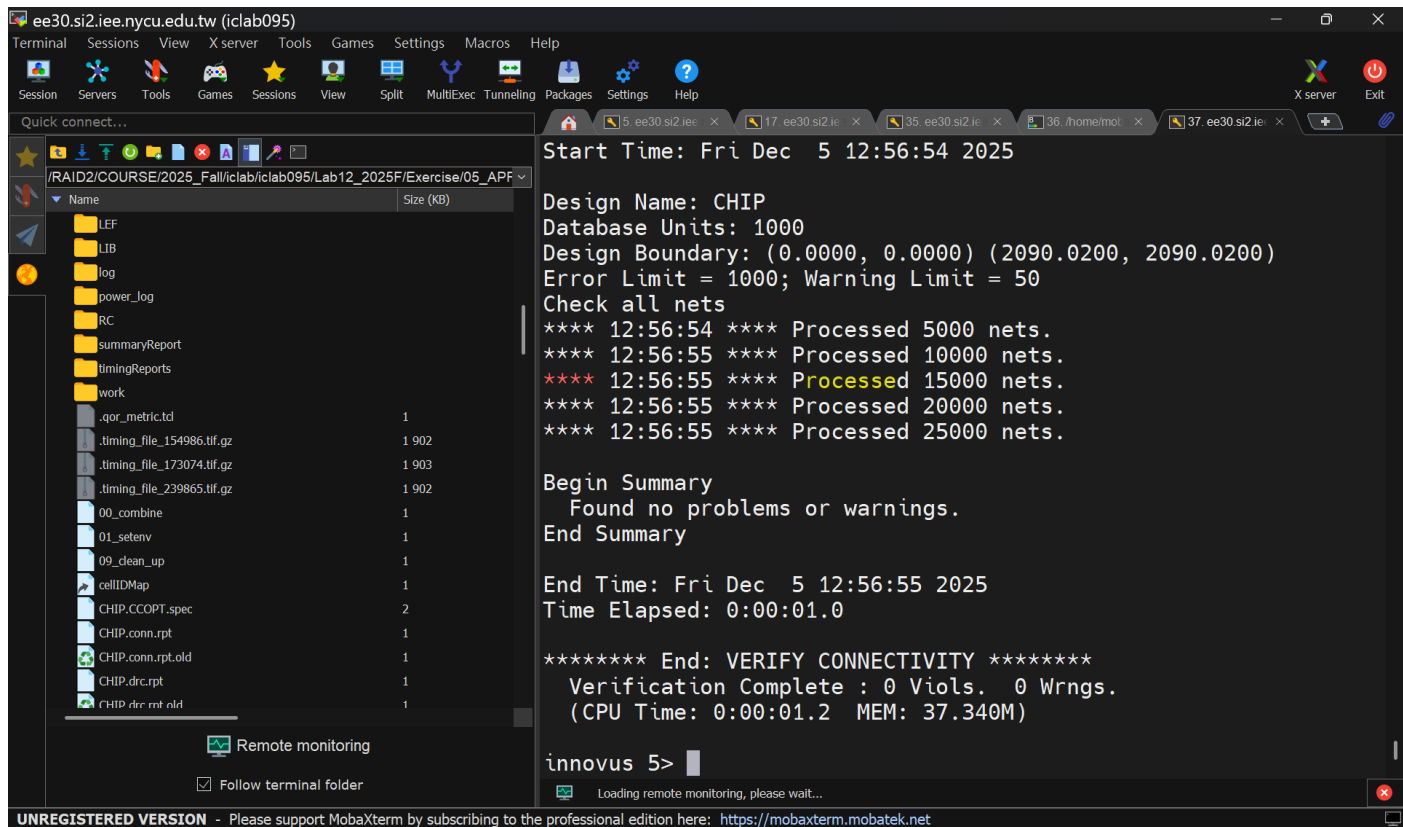
```
VERIFY DRC ..... Sub-Area : 53 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1324.800 1589.760 1589.760 1854.720} 54 o
VERIFY DRC ..... Sub-Area : 54 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1589.760 1589.760 1854.720 1854.720} 55 o
VERIFY DRC ..... Sub-Area : 55 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1854.720 1589.760 2090.020 1854.720} 56 o
VERIFY DRC ..... Sub-Area : 56 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1854.720 264.960 2090.020} 57 of 64
VERIFY DRC ..... Sub-Area : 57 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {264.960 1854.720 529.920 2090.020} 58 of
VERIFY DRC ..... Sub-Area : 58 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {529.920 1854.720 794.880 2090.020} 59 of
VERIFY DRC ..... Sub-Area : 59 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {794.880 1854.720 1059.840 2090.020} 60 of
VERIFY DRC ..... Sub-Area : 60 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1059.840 1854.720 1324.800 2090.020} 61 o
VERIFY DRC ..... Sub-Area : 61 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1324.800 1854.720 1589.760 2090.020} 62 o
VERIFY DRC ..... Sub-Area : 62 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1589.760 1854.720 1854.720 2090.020} 63 o
VERIFY DRC ..... Sub-Area : 63 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1854.720 1854.720 2090.020 2090.020} 64 o
VERIFY DRC ..... Sub-Area : 64 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:04.7 ELAPSED TIME: 5.00 MEM: 49.0M) ***
innovus 5>
```

The bottom of the terminal window shows the status "UNREGISTERED VERSION" and a link to the professional edition of MobaXterm.

7. LVS result



The screenshot shows a MobaXterm terminal window with a file explorer on the left and a terminal output on the right. The file explorer shows a directory structure for a project named 'CHIP'. The terminal output displays the start time, design name, database units, design boundary, error limit, warning limit, and a list of processed nets. It also shows a summary of the design, indicating no problems or warnings were found. The end time and time elapsed are also displayed.

```
Start Time: Fri Dec 5 12:56:54 2025

Design Name: CHIP
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (2090.0200, 2090.0200)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 12:56:54 **** Processed 5000 nets.
**** 12:56:55 **** Processed 10000 nets.
**** 12:56:55 **** Processed 15000 nets.
**** 12:56:55 **** Processed 20000 nets.
**** 12:56:55 **** Processed 25000 nets.

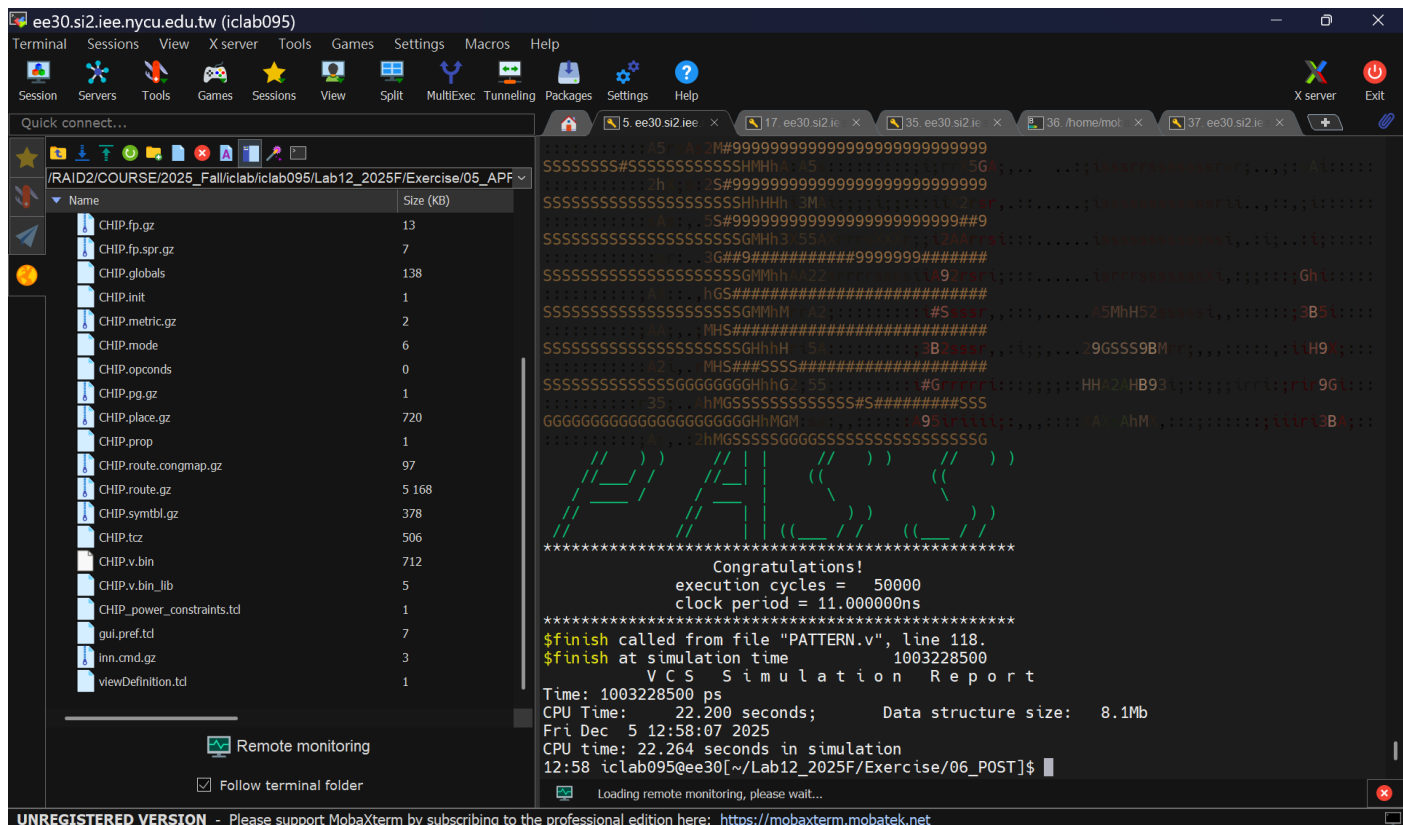
Begin Summary
Found no problems or warnings.
End Summary

End Time: Fri Dec 5 12:56:55 2025
Time Elapsed: 0:00:01.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:01.2 MEM: 37.340M)

innovus 5>
```

8. Post Layout simulation result



The screenshot shows a MobaXterm terminal window with a file explorer on the left and a terminal output on the right. The file explorer shows a directory structure for a project named 'CHIP'. The terminal output displays the start time, design name, database units, design boundary, error limit, warning limit, and a list of processed nets. It also shows a summary of the design, indicating no problems or warnings were found. The end time and time elapsed are also displayed.

```
Start Time: Fri Dec 5 12:56:54 2025

Design Name: CHIP
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (2090.0200, 2090.0200)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 12:56:54 **** Processed 5000 nets.
**** 12:56:55 **** Processed 10000 nets.
**** 12:56:55 **** Processed 15000 nets.
**** 12:56:55 **** Processed 20000 nets.
**** 12:56:55 **** Processed 25000 nets.

Begin Summary
Found no problems or warnings.
End Summary

End Time: Fri Dec 5 12:56:55 2025
Time Elapsed: 0:00:01.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:01.2 MEM: 37.340M)

innovus 5>
```


9. Power result

The screenshot shows a MobaXterm terminal window with the title "ee30.si2.iee.nycu.tw (iclab095)". The terminal displays the output of a power analysis tool. The left pane shows a file explorer with a directory structure including "power_log". The right pane shows the following text:

```
Begin Processing User Attributes
Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2388.34MB/5095.14MB/2588.25MB)

Ended Power Analysis: (cpu=0:00:33, real=0:03:00, mem(process/total/peak)=2388.34MB/5095.14MB/2588.25MB)

Begin Boundary Leakage Calculation
Ended Boundary Leakage Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2388.34MB/5095.14MB/2588.25MB)
Begin Static Power Report Generation
48 instances have no static power
*

Total Power
-----
Total Internal Power:      3.80454405      50.7252%
Total Switching Power:    3.67256332      48.9656%
Total Leakage Power:      0.02319268      0.3092%
Total Power:              7.50030006

-----
Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2390.03MB/5095.14MB/2588.25MB)

Begin Creating Binary Database
Ended Creating Binary Database: (cpu=0:00:01, real=0:00:01, mem(process/total/peak)=2904.04MB/5865.56MB/2904.04MB)

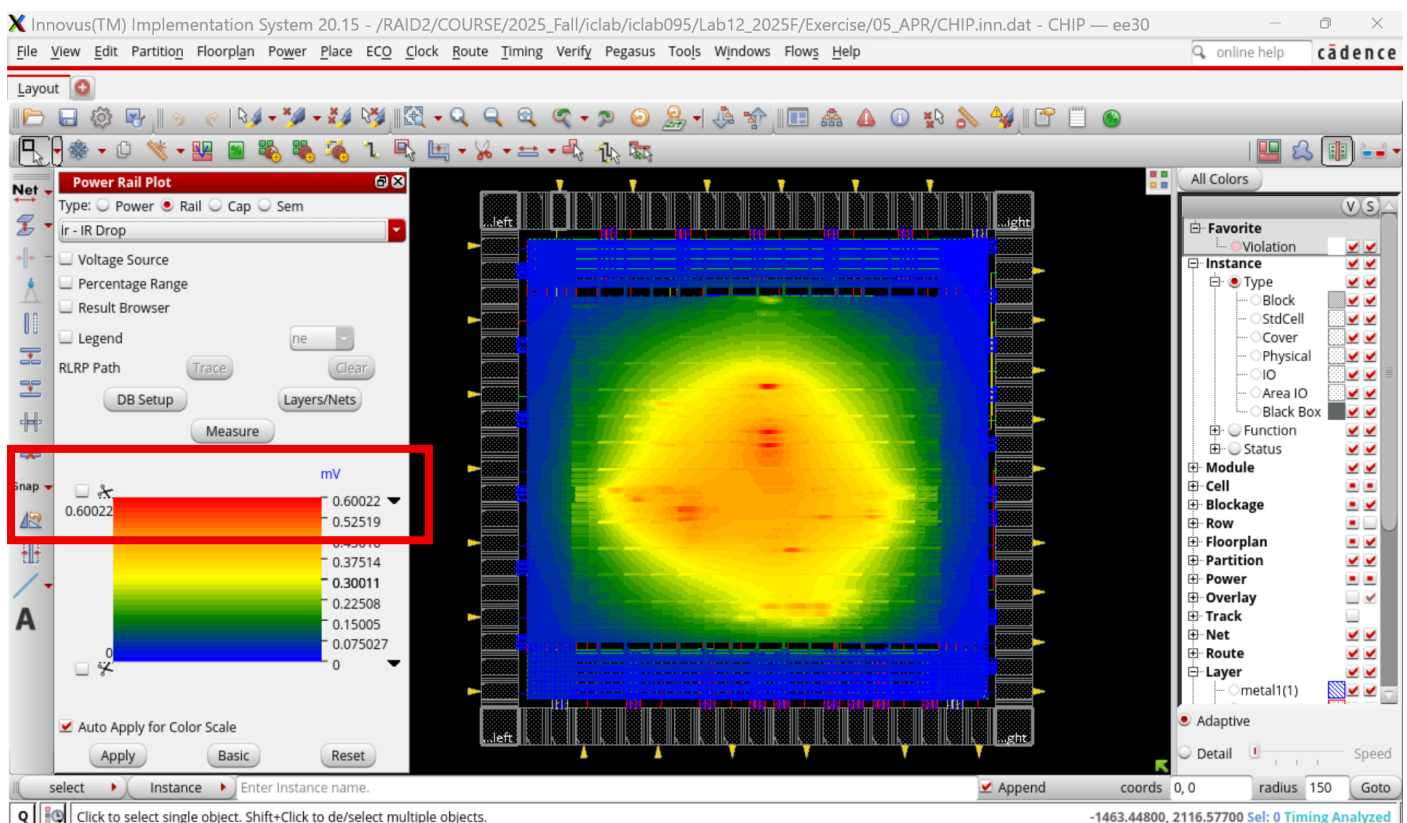
Output file is power_log/CHIP.rpt

Loading remote monitoring, please wait...
```

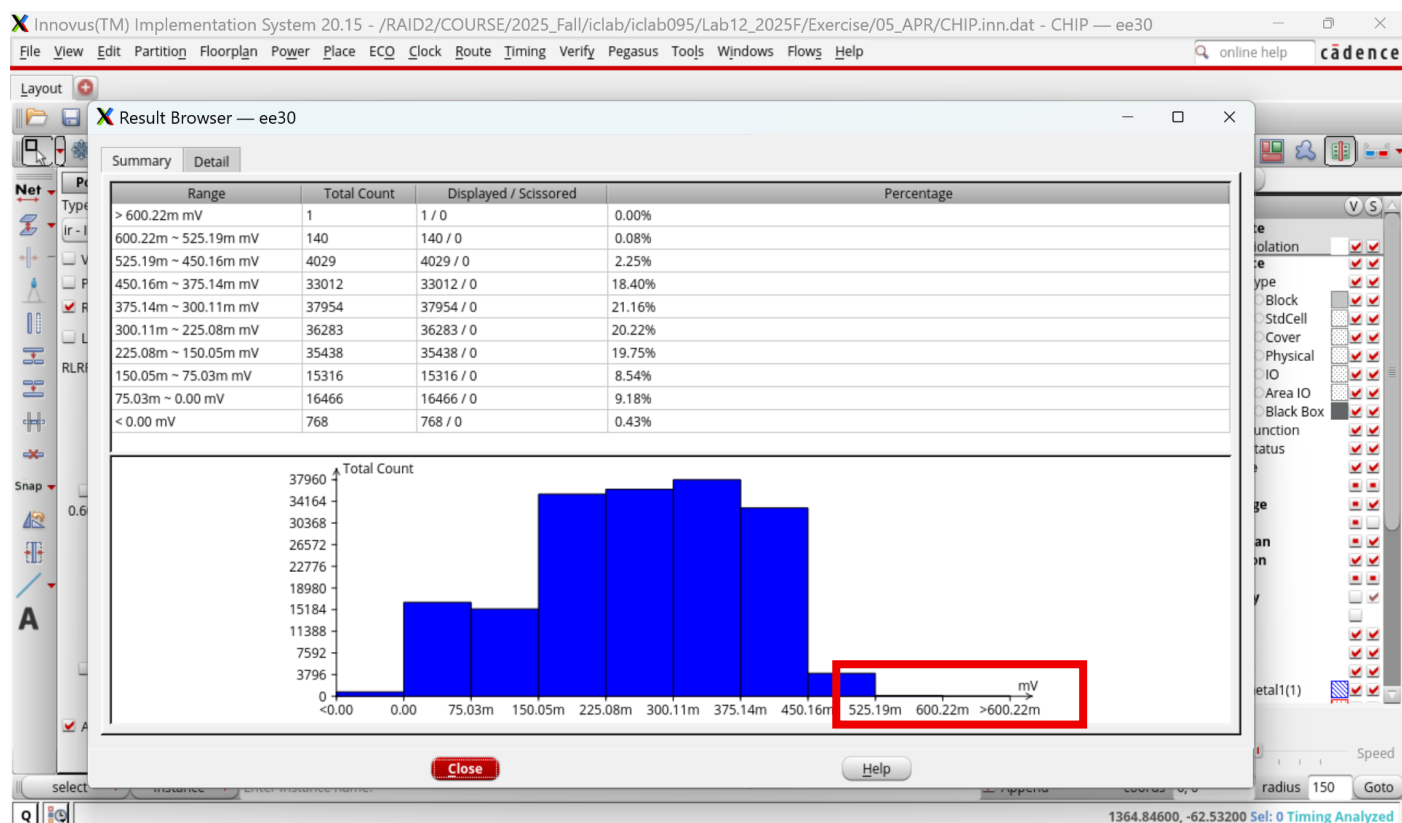
At the bottom of the terminal window, it says "UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>".

10. IR Drop Results :

maximum value of the IR drop: 0.60022mV



maximum value of the IR drop: 0.60022mV



Explanation:

目前的作法是依照助教的建議，增加 Core PowerPad 與 IO PowerPad 的數量。其中，IO PowerPad 共配置 8 組，而 Core PowerPad 則配置 16 組。

在 PowerPad 的擺放上，我也盡量依助教所述，避免將 PowerPad 集中於晶片的單一邊，而是盡可能均勻分佈於四周，以避免造成某些區域的邏輯閘因電源距離過遠而產生不利影響。

此外，目前的 power stripe 寬度設定為 4。本來有考慮將 stripe 與 stripe 之間間距（原為 100）再縮小，但由於目前的 IR Drop 檢查已通過，因此暫時尚未調整。

若後續需要再優化，則會考慮進一步增加 PowerPad 數量，或縮小 stripe 間距，不過此部分仍需視 DRC 是否能通過而定。